A Decomposition-Based Multiobjective Clonal Selection Algorithm for Hyperspectral Image Feature Selection

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***Abstract*— Feature selection is an effective way to handle the strong correlation of hyperspectral image data by screening the significant features and is generally accepted to be a multiobjective optimization problem. Nevertheless, due to the randomness of the strategies and the ambiguity of the optimization directions, the existing multiobjective evolutionary optimization-based feature selection methods can suffer from inefficient search and loss of search space with promising solutions when faced with the high-dimensional and multipeak search space. The multiobjective evolutionary algorithm based on decomposition (MOEA/D) employs a decomposition framework to provide exact guidance for the optimization directions. Unfortunately, random operators are still used, leading to inadequate local optimization. Thus, evolutionary strategies with search preference, such as clonal selection, may be necessary for local search. In this article, a novel decomposition-based multiobjective clonal selection algorithm for feature selection (MOCSA/D_FS) is proposed to obtain a feature subset with a superior classification performance. In MOCSA/D_FS, the information entropy and the ratio of the relative scatter value and mutual information are utilized as two objective functions to evaluate the information amount and redundancy. A series of subproblems are then obtained by decomposing the multiobjective problem through weight vectors with an l_2-norm constraint used to balance the search space. Subsequently, a clonal selection method with search space preference performs a detailed local search on each subproblem, which can fully exploit the potential optimal space. The effectiveness and generalizability of the proposed method were confirmed by experiments on four hyperspectral remote sensing image datasets.**

***Index Terms*— Clonal selection, feature selection, hyperspectral imagery, l_2-norm constraint, multiobjective optimization, remote sensing.**

I. Introduction

HYPERSPECTRAL remote sensing images show superior continuity in the spectral dimension and can demonstrate the reflection characteristics of targets well. This allows more detailed object identification to be achieved, contributing to the application across many fields [1], such as natural resource monitoring, disaster emergency response, and national defense construction. However, the continuous spectral information inevitably enhances the correlation and redundancy between features, and the Hughes phenomenon is triggered, which restricts the subsequent application of hyperspectral remote sensing images [2]. Feature selection can serve as an effective way to deal with this problem, whose main idea is to screen a distinctive subset from the high-dimensional features [3]. A distinctive subset is normally chosen by the use of informativeness, redundancy, and category separability to retain the critical information and remove redundancy [4].

Therefore, the problem becomes how to select an ideal subset from a much larger feature search space. The feature selection methods can be divided into three categories in terms of the evaluation approaches [5]: 1) wrapper-based methods [6], [7], where the classification accuracy is used directly as the evaluation criterion; 2) filter-based methods [8], [9], [10], where the relevant statistical characteristic parameters of the subset are utilized to form the objective functions; and 3) embedded-based methods [11], [12], where the feature selection process is directly integrated into the subsequent learning model. However, the traditional hyperspectral remote sensing image feature selection methods only consider the dominant characteristics of a single feature while not fully considering the combined characteristics of the feature subset [13]. In fact, feature selection is a complex combination optimization problem. As the number of features increases, the problem becomes high-dimensional, meaning that the traditional optimization methods based on gradients are more likely to fall into local optima.

As population-based methods, the evolutionary computation methods show a strong global optimization ability, in addition to flexible expression and relaxed application conditions, to help to find the global optimal solution in feature selection [14], [15]. The evolutionary feature selection methods based on a single evaluation criterion have been studied for a long time [16]. The basic idea is to construct an effective model to express the feature selection problem and its solutions as a population. Then, the population is optimized by evolutionary operators, and the individual with the highest fitness will serve as the final solution. From the basic idea, various evolutionary computation paradigm algorithms, such as genetic algorithms (GAs) [17], particle swarm

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optimization (PSO) algorithms [18], artificial bee colony (ABC) algorithms [19], and clonal selection algorithms (CSAs) [20], have been improved for application to feature selection. Nevertheless, due to the limitation of the single measure, the early application of the above algorithms was mainly based on the wrapper method. The main concern is that a single parameter or a combination of parameters cannot correspond directly to the subsequent application accuracy, leading to the loss of promising solutions [21], [22]. However, the high computational cost of the wrapper-based methods is increasingly unacceptable for practical applications.

Taking the above two concerns into consideration, evolutionary multiobjective feature selection methods have been proposed, which apply multiple evaluation measures to obtain a subset with a maximum amount of information, minimum redundancy, and maximum relevance within an acceptable period of time. For example, Zhang et al. [23] adopted a multiobjective immune algorithm to optimize the information entropy and cross entropy of the feature subset. Yong et al. [24] proposed a multiobjective ABC algorithm to obtain an ideal feature subset via maximizing the dispersion coefficient and minimizing the cross correlation. Wan et al. [25] proposed a multiobjective sine–cosine algorithm (SCA) based on variance and the ratio between the Jeffries–Matusita distance and mutual information (MI) for feature selection. The above multiobjective methods use multiple evaluation criteria and different search strategies to obtain a Pareto solution set, but there are still defects, such as the uneven distribution of the solution set and the insufficient local search capabilities [26]. In the face of the increasingly high-dimensional feature selection problem, exploring a Pareto solution set with more distribution diversity and local superiority puts forward higher requirements for the multiobjective evolutionary algorithms.

The multiobjective evolutionary algorithm based on decomposition (MOEA/D) is a novel optimization framework developed by Zhang and Li [27]. MOEA/D converts the multiobjective optimization problem (MOP) into a series of single-objective optimization subproblems. This method obtains a more global Pareto solution set than other traditional Pareto sorting-based methods since each subproblem represents a single optimization direction. Several studies have now proven that MOEA/D is an effective way to solve scalar optimization problems [28]. However, the direct application of MOEA/D to hyperspectral feature selection can lead to some problems [29], [30]. On the one hand, the traditional decomposition weight vectors are uniformly generated with l_1-norm constraint, but their corresponding subproblems have uneven distributions. Furthermore, the optimization of each subproblem depends on the iteration of an individual, resulting in an imbalanced search pressure and the possible loss of some dominant search space. Besides, the range mismatch of the objective functions can worsen the uneven decomposition. On the other hand, evolutionary operators based on crossover and mutation have strong randomness. Therefore, it is difficult to ensure an adequate local search within the limited iterations, causing insufficient optimization for the subproblems. Differing from random operators, the CSA adopts a strategy where individuals with high affinities are cloned at a high probability and on a large scale, meaning more detailed exploitation in the promising search space.

In this article, to address the above concerns, a novel decomposition-based multiobjective clonal selection algorithm for feature selection (MOCSA/D_FS) is proposed. The main contributions are summarized as follows.

1) Novel Framework for Multiobjective Feature Selection: In the framework, hyperspectral image feature selection is modeled as an MOP. An effective metric—the ratio between the relative scatter value (J) and the MI—is designed to maintain spectral separation and remove redundancy, together with another metric—self-information (SI)—to retain the critical characteristics. The problem is then optimized by a novel decomposition-based multiobjective CSA to output an optimal solution set. Finally, the optimal solution is automatically chosen by a knee-point selection strategy.

2) Normalized Tchebycheff Decomposition Based on Weight Vectors With l_2-Norm Constraint: For the decomposition method, the l_2-norm constraint is utilized for the generation of weight vectors so that the corresponding subproblems obtained will distribute uniformly in the objective space. Furthermore, a fairer assignment of search pressure will promote the quality of the feature subset solution to each subproblem. Normalization is implemented to eliminate the different ranges of the objective functions for comprehensive optimization.

3) Clonal Selection Method for Subproblem Neighborhood Evolution: For the evolutionary strategy of the subproblem neighborhoods, a more detailed local search is achieved through clonal amplification, hypermutation, and selection, instead of crossover and mutation. Clonal amplification based on affinity provides more chances to exploit a promising objective space with potential solutions, while hypermutation is an effective way to search for better solutions in the neighborhood of the decision space. Therefore, the clonal selection method can improve the optimization degree of the subproblems.

The rest of this article is organized as follows. In Section II, the existing related works, an overview of multiobjective optimization, and the basic methods, shortcomings, and improvement measures of MOEA/D are reviewed. The details of the framework of the proposed MOCSA/D_FS approach are described in Section III. The related experiments and analyses for the degree of Pareto optimization, the classification performance, and the parameter sensitivity are presented in Section IV. Section V concludes this article.

II. Related Research Background

In this section, the measures and improvements of the MOEA/D are reviewed and analyzed to summarize the drawbacks encountered when faced with a hyperspectral image feature selection problem. The main idea of the CSA is then introduced to handle the problem.

A. Decomposition-Based MOEA

The MOEA/D is an effective way to obtain a Pareto-nondominated solution set for an MOP [31], which is defined

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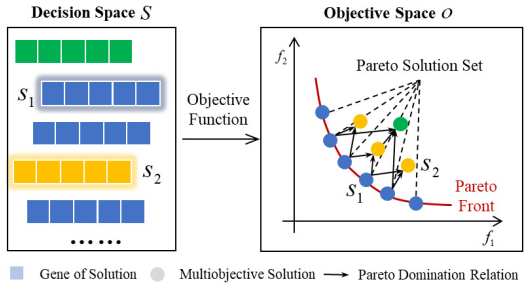


Fig. 1. Related definitions for the MOP and its solutions.

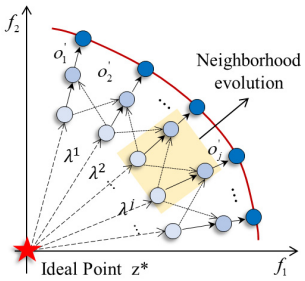


Fig. 2. Main idea of MOEA/D.

in (1), taking the minimization problem as an example

$$\begin{aligned} s^* &= \arg\min_{s\in S} \mathbf{F}(s) = \arg\min_{s\in S}(f_1(s), f_2(s), \ldots, f_m(s)) \\ g_i(s) &\le 0 \quad i = 1, 2, \ldots, p \\ h_j(s) &= 0 \quad j = 1, 2, \ldots, q \end{aligned} \tag{1}$$

where s represents the solution from the decision space S, $f(s)$ refers to the m objective functions, and $g(s)$ and $h(s)$ are the p inequality and q equality constraints in the MOP. There are always contradictory choices in an MOP, i.e., the multiple objective functions contradict each other and require tradeoffs, especially in the fields of engineering, biology, and economics.

The solution for the MOP often reaches the optimum on one objective, and the performance for the others will reduce significantly. Therefore, the Pareto domination relationship has been introduced into the MOP. For the two solutions shown in (2), s_1 dominates s_2, which can be denoted as $s_1 \succ s_2$

$$\begin{aligned} &\forall i \in \{1, 2, \ldots, m\}, f_i(s_1) \le f_i(s_2) \\ &\quad \cap \exists i \in \{1, 2, \ldots, m\}, f_i(s_1) < f_i(s_2). \end{aligned} \tag{2}$$

All the Pareto solutions are combined to appear as Pareto-optimal solution sets in the decision space and form an optimization surface in the objective space known as the Pareto front (PF). Fig. 1 expresses the related definitions for the MOP and its solutions.

The question of how to approximate the PF then arises. The MOEA/D algorithm provides a framework for obtaining the PF by decomposing the MOP through weight vectors, which is given as follows:

$$\min_{\mathbf{s}\in S} g^{\text{tch}}\left(\mathbf{F}(\mathbf{s})|\boldsymbol{\omega}, \mathbf{z}^*\right) = \max_{1\le i\le m}\left\{w_i|f_i(\mathbf{s}) - z_i^*|\right\} \tag{3}$$

where $g^{\text{tch}}(o)$ represents the subproblem generated by the weight vector $\boldsymbol{\omega} = (w_1, w_2, \ldots, w_m)$ with $\sum_{i=1}^m w_i = 1$ and $w_i > 0$, and $\mathbf{z}^* = (z_1^*, \ldots, z_m^*)$ with $z_i^* = \min(f_i(\mathbf{s}))$ is the ideal point in the objective space. It is clear that different weight vectors represent different degrees of multiobjective tradeoffs and also divide the entire optimized objective space into multiple subspaces, as shown in Fig. 2. The subproblem objective function corresponding to each weight vector has a certain geometric distribution property. The MOEA/D algorithm continuously updates the ideal point and subproblem optimal solution by utilizing crossover and mutation methods to advance the optimization of the entire PF [27], [32].

However, related studies have shown that the subproblems obtained by the traditional decomposition method do not have uniform search regions, which can result in insufficient local exploitation for MOPs [33]. To handle the adverse geometric properties of the subproblem distribution and conduct ample search in a wide space, Deb and Jain [34] developed a modified Tchebycheff decomposition approach, which adopts division between $f_i(\mathbf{s}) - z_i^*$ and λ_i instead of multiplication, for which the calculation method is shown as follows:

$$\min_{\mathbf{s}\in S} g^{\text{mtch}}\left(\mathbf{F}(\mathbf{s})|\boldsymbol{\omega}, \mathbf{z}^*\right) = \max_{1\le i\le m}\left\{\frac{|f_i(\mathbf{s}) - z_i^*|}{w_i}\right\} \tag{4}$$

with the weight vector $\boldsymbol{\omega}$ with $\sum_{i=1}^m w_i = 1$ and $w_i > 0$. This decomposition approach makes the search space of the subproblems consistent with the weight vectors. However, in fact, in the case of a symmetrical distribution of the weight vectors along the diagonal of the objective space, this method is essentially equivalent to the traditional one.

Therefore, in order to study the geometric characteristics of the weight vectors and then highlight the preferred search area for detailed exploitation, Ma et al. [35] put forward a novel Tchebycheff decomposition method for weight vectors based on the l_p-norm constraint, as shown in the following equation:

$$\min_{\mathbf{s}\in S} g^{\text{ptch}}\left(\mathbf{F}(\mathbf{s})|\boldsymbol{\lambda}, \mathbf{z}^*\right) = \max_{1\le i\le m}\left\{\frac{|f_i(\mathbf{s}) - z_i^*|}{\lambda_i}\right\} \tag{5}$$

with the weight vector $\boldsymbol{\lambda}$ with $\|\lambda_i\|_p = 1$ and $\lambda_i > 0$. The preferred search regions of the corresponding subproblems vary with the different values of p, and a uniform search area can be obtained when p equals 2. In hyperspectral image feature selection, it is generally accepted that there is no preferred region. Therefore, the l_2-norm constraint is utilized to balance the search pressure between each individual, under the premise of global capabilities. In addition, the traditional random evolutionary strategy has been improved to a strategy based on clonal selection to gain stronger local capabilities as follows.

B. Clonal Selection Algorithm

The CSA was modeled by De Castro and Von Zuben [36] in 2000 based on the principle of the human immune system. The biological immune system is a defense mechanism that has evolved to protect the body from "nonself" aggressions known as antigens. The immune system is stimulated by the antigen to produce B cells and generate antibodies, which can

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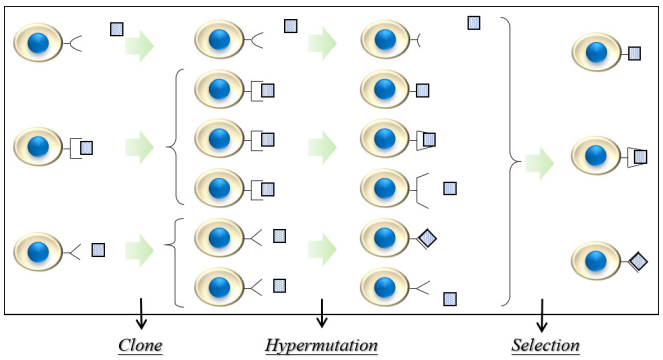


Fig. 3. Principle of clonal selection.

specifically bind to antigens by complementary matching. The strength of the binding, which is called "affinity," depends on the pattern matching.

In the process of clonal selection in the biological immune system, antibodies with high affinity with antigens are selected. The corresponding B cells then rapidly proliferate and differentiate through cloning and hypermutation, generating optimal antibodies and, eventually, eliminating the antigens. Some high-affinity B cells are converted into memory cells that react quickly when attacked by the same or similar antigen. The principle is shown in Fig. 3.

By simulating the above process, the CSA has been applied to many engineering problems, such as task scheduling [37] and network design [38]. The pseudocode of the algorithm is given in Algorithm 1.

Algorithm 1 Pseudocode of the CSA

Input:

1) PS: the size of population;
2) T_{max}: the maximum number of iterations;
3) K: the number of solutions involved in the clone;

Output:

1) the best solution $\boldsymbol{s}^*$.

initialize *a population* $\boldsymbol{P_0}$ *with* $\boldsymbol{F}(\boldsymbol{s}_i^0)$, $\boldsymbol{s}_i^0 \in \boldsymbol{P_0}$;
for $t \leftarrow 1$ ***to*** T_{max} ***do***
  $\boldsymbol{P_{sel}} \leftarrow$Select($\boldsymbol{P_{t-1}}$, K);
  $\boldsymbol{C_t} \leftarrow$ Clone($\boldsymbol{P_{sel}}$, PS);
  $\boldsymbol{P_t} \leftarrow$ Hypermutate($\boldsymbol{C_t}$);
end
return $\boldsymbol{s}^* \leftarrow \boldsymbol{s}_j^{T_{\max}}$, $j = \underset{i=1,2,\ldots,PS}{\arg\max} \{\boldsymbol{F}(\boldsymbol{s}_i^{T_{\max}})\}$.

Usually, the problem to be solved is abstracted as an antigen, with its solutions as B cells. The solutions form the population $\boldsymbol{P_0}$ by encoding. By evaluating the objective functions between them, the more suitable the solutions are, the more likely they are to be chosen and cloned on a larger scale to constitute population $\boldsymbol{C_t}$. In this way, the promising search space with potential solutions will be more likely to be exploited carefully, which can be considered as a preference for search space. After this, the cloned solutions from $\boldsymbol{C_t}$ undergo minor corrections known as hypermutation to find the optimal solution within the neighborhood. In addition, strategies such as antibody editing of cloned populations and updating of inferior solutions can also be used to enhance the globality of the solutions. The main iterative process of clonal selection is completed as described above. The solution that best fits the problem can be obtained after many iterations.

Compared with the traditional GAs, the CSA employs an evolutionary strategy with search preference. This preference is reflected in different spatial search intensities according to the affinity and the use of microspace mutation strategies for detailed development. Therefore, the clonal selection strategy has more advantages over the crossover and mutation strategies with strong randomness.

III. Proposed MOCSA/D_FS Method

Due to the imbalance of the search space of the subproblems obtained by the traditional Tchebycheff decomposition method and the search inadequacy of the random evolutionary strategies, a normalized l_2-norm constraint decomposition method-based multiobjective algorithm is proposed. The idea of clonal selection is employed in the neighborhood evolution of the subproblems for the tradeoff selection among the hyperspectral features between high information amount and low redundancy of the subset. Two efficient objective functions are, thus, established and continuously optimized over the course of the evolution. The main methodological framework of MOCSA/D_FS is shown in Fig. 4.

A. Multiobjective Modeling for MOCSA/D_FS

The ideal solution for the feature selection problem should achieve a tradeoff in the following aspects: critical information retention and redundant information removal. The problem can be modeled as an MOP by utilizing the appropriate parameters and their combinations to form objective functions.

In this article, two criteria of the characteristics of feature subsets are constructed related to information theory. One of the objective functions is the mean of the information entropy, which is denoted as SI (6). Entropy measures the degree of chaos in a material system, while information entropy quantifies information by describing the uncertainty of the occurrence of an event. For imagery, information entropy can be calculated with the occurrence probability of normalized pixel values. Therefore, the SI reflects the average discreteness of pixels, thereby characterizing the information amount. The relevant formulas are given as follows:

$$SI = \frac{1}{N_B}\sum_{i=1}^{N_B} H(B_i) = \frac{1}{N_B}\sum_{i=1}^{N_B}\left(-\sum_{x\in B_i} p(x)\cdot\log(p(x))\right) \tag{6}$$

where N_B represents the size of the selected feature subset, $H(B_i)$ refers to the information entropy of the ith feature image, and $p(x)$ denotes the probability of the pixel gray value being x, which can be parsed in (7) at a deeper level

$$p(x) = \frac{\text{hist}(x)}{M \times N} \tag{7}$$

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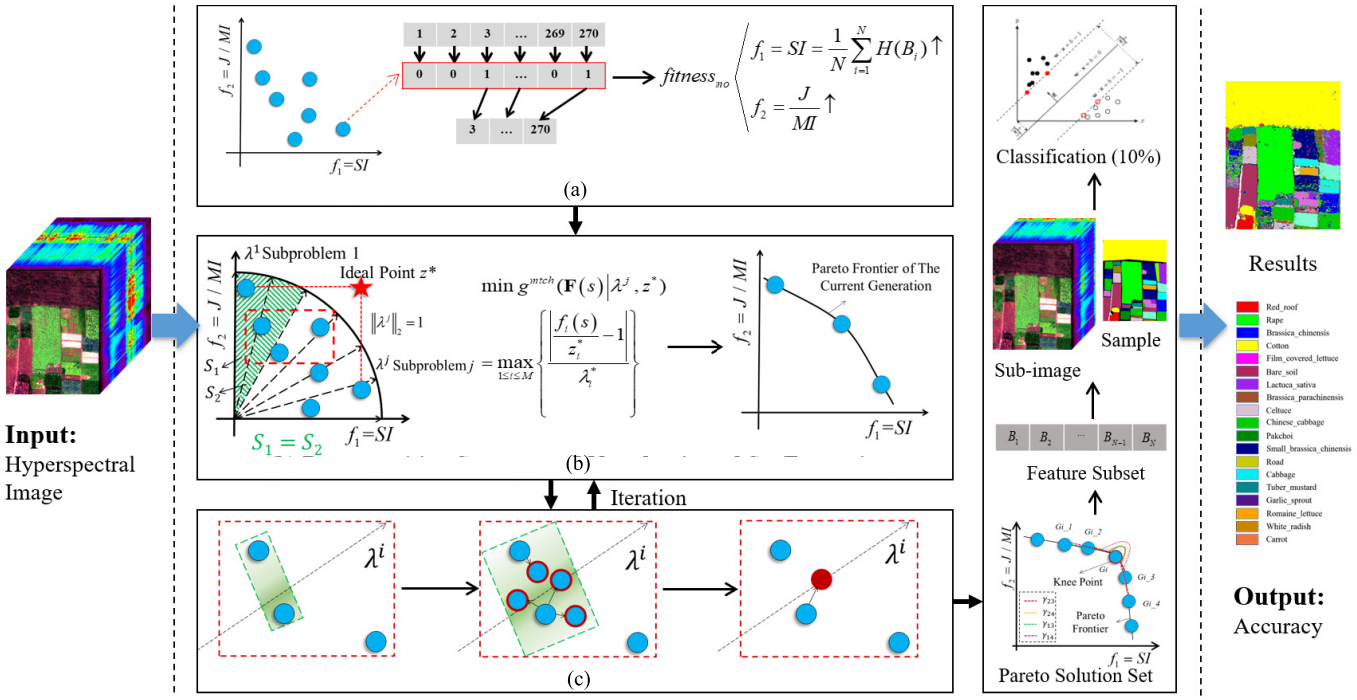


Fig. 4. Main methodological framework of MOCSA/D_FS. (a) Population initialization and fitness calculation. (b) Decomposition strategy and nondominated set extraction. (c) Clonal selection and subproblems update.

where hist(x) is the histogram and $M \times N$ is the total number of pixels.

The other is the ratio between the scatter matrix parameters, which are defined as J (8) [39], and the average MI, which is known as the MI (11). J is the trace of the ratio result of the interclass scatter matrix (9) to the intraclass scatter matrix (10), which guarantees a more class-separated subset by maximizing the interclass distances. Moreover, the MI is another vital index to assess redundant information between different features from the information theory perspective. Specifically, the joint probability can be calculated through the joint histogram of the feature image pixel gray values and then converted into the MI between two features. Therefore, the smaller the ratio of J and MI, the better the category divisibility and redundancy of the selected subset. The related formulas are given as follows:

$$J = \text{trace}\left(S_w^{-1} \cdot S_b\right) \tag{8}$$

where S_w and S_b are known as the intraclass scatter matrix and interclass scatter matrix, respectively. Their calculation methods are given in the following:

$$S_w = \sum_{i=1}^{N_c} P(\omega_i) \cdot \text{cov}(\omega_i) \tag{9}$$

$$S_b = \sum_{i=1}^{N_c} P(\omega_i)(\mu_i - \mu_0)(\mu_i - \mu_0)^T \tag{10}$$

where N_c is the total number of classes in the image, and $P(\omega_i)$ and $\text{cov}(\omega_i)$, respectively, represent the prior probability and the covariance matrix of the ith class. μ_i refers to the categorical mean vector, while μ_0 is the mean of μ_i in all the classes. Naturally, the parameter J, which is a combination of these two matrices, can describe category separability.

According to the information theory, the MI between two feature images is defined as follows:

$$\begin{aligned} MI &= \frac{1}{C_{N_B}^2} \sum_{i=1}^{N_B-1} \sum_{j=i+1}^{N_B} I\left(B_i; B_j\right) \\ &= \frac{1}{C_{N_B}^2} \sum_{i=1}^{N_B-1} \sum_{j=i+1}^{N_B} \left(\sum_{x \in B_i} \sum_{y \in B_j} p(x,y) \log \frac{p(x,y)}{p(x) \cdot p(y)} \right) \end{aligned} \tag{11}$$

in which $C_{N_B}^2$ represents the combination number of selecting two from N_B features. All the features are combined and calculated in pairs, and then, the average value is the final result. $I(B_i; B_j)$ relies on the joint probability between two features and the respective probability distribution. Statistically, the joint probability can be boiled down to a 2-D histogram.

Overall, feature selection is modeled as an MOP (13), where the SI helps to retain features with key information, and the J/MI ratio prefers ones with low redundancy

$$\max \mathbf{F}(s) = \max\{f_1, f_2\} \tag{12}$$

$$\begin{cases} f_1 = \text{SI} \\ f_2 = \frac{J}{\text{MI}}. \end{cases} \tag{13}$$

The encoding method of the solutions is also a key component of the model. Since each feature has only two states of choice or not, it is reasonable to use binary encoding. In this case, the code length is the number of features, causing the search space to expand exponentially.

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B. Framework of MOCSA/D_FS

In this part, based on the multiobjective optimization model and the encoding method of feature selection, the implementation process of MOCSA/D_FS is analyzed and detailed in individual steps, as follows. The pseudocode is presented in Algorithm 2.

Algorithm 2 Pseudocode of MOCSA/D_FS

Input:

1) *img*: hyperspectral image dataset with labels;
2) N_B: the preset size of selected band subset;
3) *PS*: the size of population;
4) T_{max}: the maximum number of iterations;
5) L: the size of neighborhood;

Output:

1) the best solution $\boldsymbol{s}^*$.

initialize *a population* $\boldsymbol{P_0}$ *with* $\boldsymbol{F}(\boldsymbol{s}_i^0)$, $\boldsymbol{s}_i^0 \in \boldsymbol{P_0}$ *and weight vectors* $\boldsymbol{\lambda}^i$ *by* (14);
$\boldsymbol{z}^* \leftarrow \max\{\boldsymbol{F}(\boldsymbol{s}_i^0)|\boldsymbol{s}_i^0 \in \boldsymbol{P_0}\}$;
$g_{i,j}^{2ntch} \leftarrow g^{2ntch}(\boldsymbol{F}(\boldsymbol{s}_j)|\boldsymbol{\lambda}^i, \boldsymbol{z}^*)$;
for $i \leftarrow 1$ ***to*** *PS* ***do***
  $\boldsymbol{s}_i^0 \leftarrow \boldsymbol{s}_j^0, j = \underset{k=1,2,\ldots,PS}{\arg\max}\{g_{i,k}^{2ntch}\}$;
end
$\boldsymbol{EP} \leftarrow \left\{\boldsymbol{s}_i^0 \middle| \nexists \boldsymbol{s}_j^0 \in \boldsymbol{P_0}, \boldsymbol{s}_j^0 \succ \boldsymbol{s}_i^0\right\}$;
for $t \leftarrow 1$ ***to*** T_{max} ***do***
  for $i \leftarrow 1$ ***to*** *PS* ***do***
    $\boldsymbol{g}_i^{2ntch} \leftarrow \left\{g_{i,j}^{2ntch} | j \in B(i)\right\}$;
    $(\boldsymbol{\Delta}, \boldsymbol{pc}) \leftarrow \text{rankfun}(\boldsymbol{g}_i^{2ntch})$ *by* (16)–(17);
    $\boldsymbol{C}_t \leftarrow \left\{\underbrace{\boldsymbol{s}_k^{t-1}, \ldots, \boldsymbol{s}_k^{t-1}}_{n_k} | pc_k > rand_k\right\}$ *by* (18);
    if $\delta > rand$
      $\boldsymbol{TP} \leftarrow$Hypermutate($\boldsymbol{C}_t$);
    else
      $\boldsymbol{TP} \leftarrow$Hypermutate($\left\{\boldsymbol{s}_j^{t-1} | j \in B(i)\right\}$);
    end
    $\boldsymbol{z}^* \leftarrow \max\{\boldsymbol{z}^*; \boldsymbol{F}(\boldsymbol{s}_i) | \boldsymbol{s}_i \in \boldsymbol{TP}\}$;
    $\boldsymbol{EP} \leftarrow \left\{\boldsymbol{EP}/\boldsymbol{s}_j \middle| \forall \boldsymbol{s}_i \in \boldsymbol{TP}, \forall \boldsymbol{s}_j \in \boldsymbol{EP}, \boldsymbol{s}_i \succ \boldsymbol{s}_j\right\}$;
    $\boldsymbol{EP} \leftarrow \left\{\boldsymbol{EP}, \boldsymbol{s}_i \middle| \forall \boldsymbol{s}_i \in \boldsymbol{TP}, \nexists \boldsymbol{s}_j \in \boldsymbol{EP}, \boldsymbol{s}_j \succ \boldsymbol{s}_i\right\}$;
    for j ***in*** $B(i)$ ***do***
      $\boldsymbol{s}_j^t \leftarrow \boldsymbol{s}_n, n = \underset{k=1,2,\ldots,L+1}{\arg\max}\left\{\boldsymbol{F}(\boldsymbol{s}_k) \middle| \boldsymbol{s}_k \in \left\{\boldsymbol{TP}, \boldsymbol{s}_j^{t-1}\right\}\right\}$;
    end
  end
end
return *the knee point of* $\boldsymbol{EP}$.

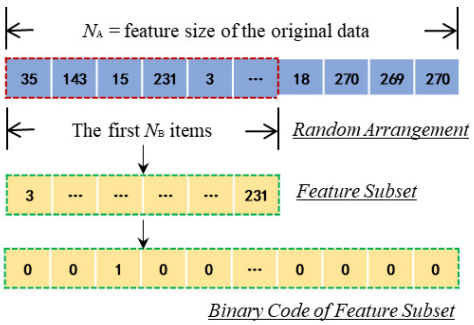


Fig. 5. Initialization method for the feature subset solution.

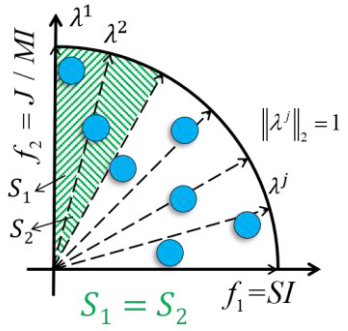


Fig. 6. Weight vectors with l_2-norm constraint.

1) Pretreatment of the Dataset: Since each feature of the input dataset may distribute over different ranges with specific physical meanings, it is necessary to normalize the data. The data are then divided by the semantic classes, and 10% of the samples consist of the training set while the rest form the test set.

2) Encoding and Initialization of the Population: According to the preset size of feature subset N_B, a random arrangement of 1 to N_A (where N_A is the total band number) is generated. The first N_B terms of the arrangement are selected as the feature position ordinals of this random solution. The binary code of the solution is obtained to serve as the manipulative carrier of the evolutionary operators, with "1" meaning that the feature is chosen and "0" meaning that the feature is not chosen. Based on this approach, Fig. 5 expresses the initialization method for the solution. In this way, a solution population of PS individuals can be obtained.

3) Determination of the References and Decomposition Strategy: The initialized population is evaluated by (13) to obtain the ideal point corresponding to the coordinates of the best fitness values of the two objective functions. The ideal point is updated instantly as the optimal fitness of the objective changes. In order to decompose an MOP into a series of subproblems, weight vectors should be determined to guide the comprehensive optimization direction. In MOCSA/D_FS, the weight vectors are generated with the normalized l_2-norm constraint in (14), as shown in Fig. 6

$$\boldsymbol{\lambda}^i = \left(\cos\left(\frac{i-1}{\text{PS}-1} \cdot \frac{\pi}{2}\right), \sin\left(\frac{i-1}{\text{PS}-1} \cdot \frac{\pi}{2}\right)\right)^T. \quad (14)$$

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As shown above, the weight vectors divide the objective space by angle and are then converted into dimensional components with the help of trigonometric functions. The corresponding subproblems can be represented by (15).

Compared with (5), the proposed Tchebycheff decomposition mainly uses the ideal point to normalize the objective vector, eliminating the effect of range inhomogeneity on the decomposition method. In addition, the neighborhoods of the subproblems are organized by the Euclidean distances of the weight vectors. For each subproblem, the first L vectors with the smallest distance are selected, their corresponding subproblems are regarded as the neighborhoods of the current subproblem, and the subsequent evolutionary strategies are performed on each small neighborhood

$$\min_{\mathbf{s}\in S} g^{\text{2ntch}}\left(\mathbf{F}(\mathbf{s})|\boldsymbol{\lambda}, \mathbf{z}^{*}\right) = \max_{1\le i\le m}\left\{\frac{|\frac{f_i(\mathbf{s})}{z_i^{*}} - 1|}{\lambda_i}\right\}. \tag{15}$$

4) Rearrangement of the Populations and Extraction of Nondominated Sets: Since the arrangement of the initial randomly generated solutions is irregular, the subproblem that each solution is most suitable for is inconsistent with its ordinal number. Therefore, assigning each solution directly to the subproblem in the default order does not result in the best fit. For each subproblem, the entire population is calculated by (15), and the individual with the smallest g^{2ntch} is assigned, avoiding the influence of initialization on the optimization.

On the other hand, there may be cases where nondominated solutions iterate each other in the process of optimization of the subproblems, and directly taking the final population output as the result may lead to the omission of nondominated solutions. Thus, the nondominated solutions generated during the entire iteration are stored through an external Pareto (EP) set.

The EP set is updated by Pareto domination analysis of each new solution with the original EP set. Suppose that the new solution is s_i^t, and the solution of the original EP set is y^t. If s_i^t is not dominated by any solution y_j^t in EP, then s_i^t will be stored, and its dominant solution y_d^t will be deleted. In contrast, s_i^t will not be stored when some solutions y_{nd}^t dominate.

5) Clonal Selection and Hypermutation: In MOCSA/D_FS, the idea of clonal selection is applied. Specifically, for each subproblem i, the solution individuals j within its neighborhood $B(i)$ are calculated by (15) according to the current weights λ^i. The proportion of cloned Δ_j is determined by the proportion of the subproblem fitness g_j^{2ntch} for each solution j according to (16). At the same time, the fitness is sorted in ascending order r_j corresponding to the probability pc_j of the solution being selected (17)

$$\Delta_j = \frac{\sum_{n=1}^{L} g_n^{\text{2ntch}} - g_j^{\text{2ntch}}}{\left(1-\frac{1}{L}\right)\cdot\sum_{n=1}^{L} g_n^{\text{2ntch}}} \tag{16}$$

$$\text{pc}_j = \frac{1-p_{\min}}{1+p_{\min}\cdot\exp\left(20\cdot\left(\frac{r_j}{L}-0.5\right)\right)} \tag{17}$$

where L represents the size of $B(i)$ and $p_{\min}$ is the preset minimum selection probability. It can be seen that the more suitable the solution in the neighborhood is for the subproblem, the higher the probability, and the larger the scale that can be cloned.

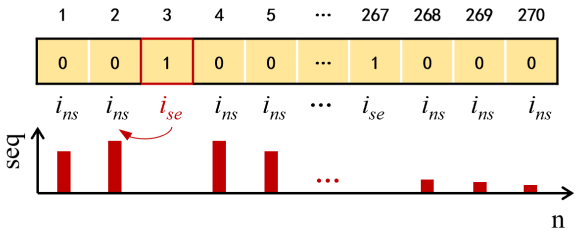


Fig. 7. Principle of hypermutation.

After this, the neighborhoods are filtered using random sequences to obtain K selected neighborhood solutions. As shown in (18), these solutions are cloned proportionally to make up clone population $\boldsymbol{C}_{\mathbf{t}}$

$$n_k = \text{round}\left(L\cdot\frac{\Delta_k}{\sum_{n=1}^{K}\Delta_n}\right), \text{pc}_k > \text{rand}_k \tag{18}$$

where n_k is defined as the number of clones of the kth selected individual. In order to enhance the robustness, MOCSA/D_FS uses the parameter δ to determine whether the population undergoing hypermutation is the clone population $\boldsymbol{C}_{\mathbf{t}}$ described above or a neighborhood population $B(i)$.

Hypermutation is a tiny mutation that occurs within the geometric neighborhood of the individual. MOCSA/D_FS adopts a new pattern of swap mutation to give full play to the local search capability of hypermutation: randomly take a selected feature i_{se}, compute the absolute distance dis between the other unselected features i_{ns} to i_{se}, and take the reciprocal as the selection intensity seq of the positions. Determine an actual location through the wheel selection method. This method ensures that the unselected features closer to the mutation location are more likely to be selected. The principle of hypermutation is demonstrated in Fig. 7.

6) Updating of the Population and External Pareto Set: For new populations obtained by hypermutation, their objective vectors are calculated first, and the ideal point $\mathbf{z}^{*}$ and the EP set are updated as described in step 4). Then, in the neighborhood of the current subproblem $B(i)$, the fitness of the new solution for each neighborhood subproblem j is calculated and compared with the original solution. The individual with the smallest fitness is retained as the solution to the corresponding subproblem. In this way, the whole subproblem set can be updated, and all the nondominated solutions are obtained through the memory set after many iterations.

7) Stopping Condition: Repeat steps 5)–6) until the maximum number of iterations is reached. Since the optimization result is a solution set, which represents different degrees of tradeoff, classifying all the solutions within a solution set clearly entails a significant computational burden. A knee-point-based method [40] is applied to extract the best solution in the PF.

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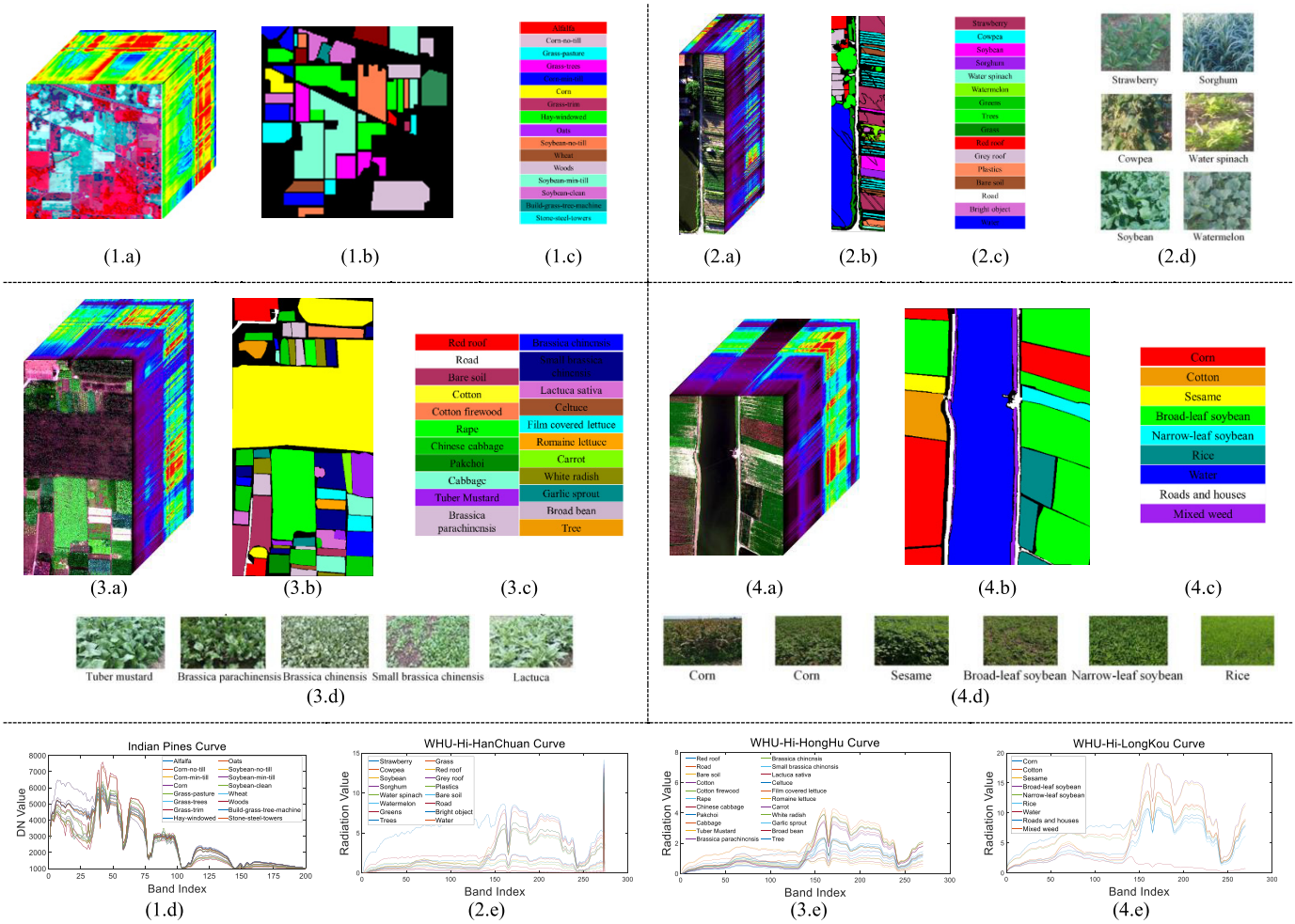


Fig. 8. Four hyperspectral remote sensing image datasets: 1) Indian Pines dataset; 2) WHU-Hi-HanChuan dataset; 3) WHU-Hi-HongHu dataset; and 4) WHU-Hi-LongKou dataset. (1.a) Image. (1.b) Ground-truth image. (1.c) Legend. (1.d) Spectral curves. (2.a) Image. (2.b) Ground-truth image. (2.c) Legend. (2.d) Some typical ground objects. (2.e) Spectral curves. (3.a) Image. (3.b) Ground-truth image. (3.c) Legend. (3.d) Some typical ground objects. (3.e) Spectral curves. (4.a) Image. (4.b) Ground-truth image. (4.c) Legend. (4.d) Some typical ground objects. (4.e) Spectral curves.

IV. Experiments and Analyses

In order to confirm the effectiveness of MOCSA/D_FS, we compared it with the following feature selection methods: the traditional nonevolutionary method of unsupervised band selection by dominant set extraction (DSEBS) [41], the optimal clustering framework for band selection (NC-OC-IE) [42], the fast neighborhood grouping method for band selection (FNGBS) [43], and the single-objective evolutionary feature selection methods under the metric of J/MI, i.e., binary versions of GA, PSO, DE, and SCA. In addition, the proposed method is compared with some typical multiobjective evolutionary methods under the metrics of SI and J/MI, i.e., nondominated sorting GA-II/-III (NSGA-II/-III) [44], [45], [46], multiobjective PSO [47], and MOEA/D [48]. The multiobjective SCA (MOSCA*) [25] for feature selection was tested under the same metrics as MOCSA/D_FS as well.

A. Experimental Datasets

In order to test the effectiveness and generalizability of MOCSA_FS, four sets of hyperspectral remote sensing image datasets were used: the Indian Pines public dataset [49] and three sets of Wuhan University unmanned aerial vehicle (UAV) datasets, i.e., WHU-Hi-HanChuan, WHU-Hi-HongHu, and WHU-Hi-LongKou [50].

Among the four hyperspectral datasets, the first one was acquired in a pure agriculture scene with certain noise by the Airborne Visible/Infrared Imaging Spectrometer at a resolution of 20 m. The others are known as H^2 datasets, which was captured by the Headwall Nano-Hyperspec sensor and contains a mixture of natural land cover and crops. The images have been denoised to some extent. As for the latter three datasets, WHU-Hi-LongKou has fewer clustered species, making it easy to interpret overall; WHU-Hi-HanChuan has many classes with a small sample and a striped distribution, hard for accurate classification; and WHU-Hi-HongHu owns the highest resolution within the test datasets, and the captured region covers different cultivars of the same crop with the similar spectrum, difficult to distinguish effectively. Fig. 8 and Table I show the related information for the datasets.

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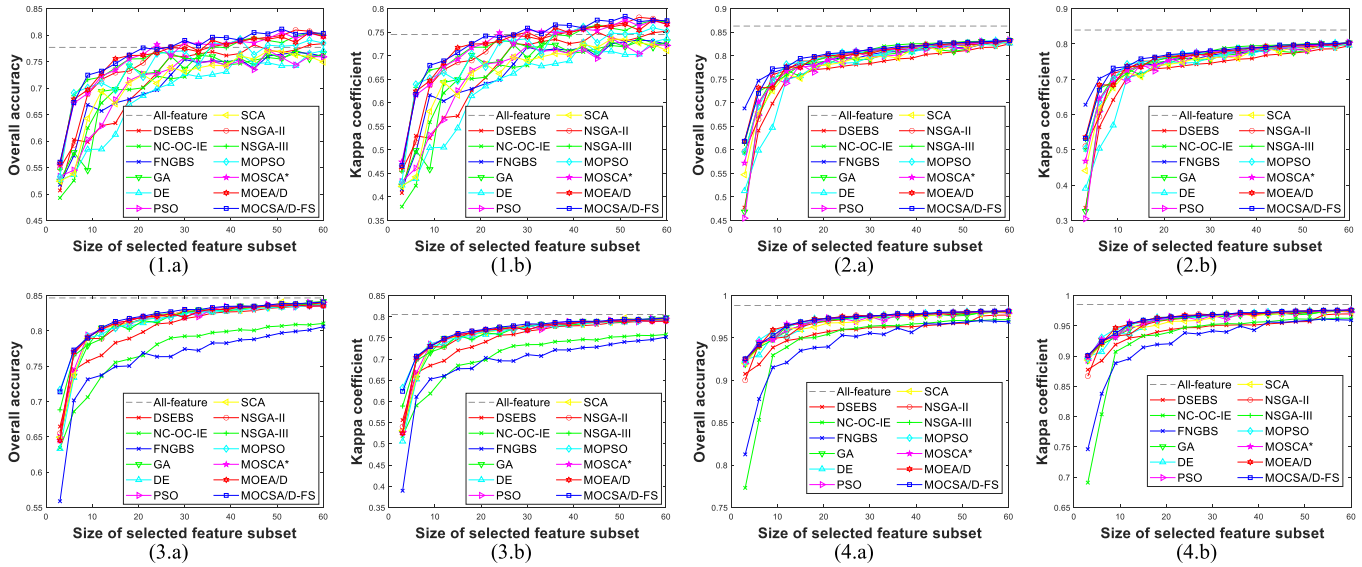


Fig. 9. Classification accuracy curves of the different methods for the four hyperspectral remote sensing image datasets. (1) Indian Pines dataset. (2) WHU-Hi-HanChuan dataset. (3) WHU-Hi-HongHu dataset. (4) WHU-Hi-LongKou dataset. (a) Overall accuracy. (b) Kappa coefficient.

TABLE I
DETAILS OF THE FOUR HYPERSPECTRAL IMAGE DATASETS

Attributes	Indian Pines	WHU-Hi-HanChuan	WHU-Hi-HongHu	WHU-Hi-LongKou
# Reso.	20 m	0.109 m	0.043 m	0.463 m
# Features	200	274	270	270
# Classes	16	16	22	9
# Size	145 × 145	303 × 1217	475 × 940	400 × 550

Notes: Reso.: spatial resolution of the hyperspectral image dataset.

From the descriptions of the four hyperspectral remote sensing image datasets, the following vital characteristics can be seen: 1) the large numbers of features for the datasets mean that the dimension of the feature space is very high, difficult for model interpretation; 2) so many categories in the datasets mean that the misclassification among categories will increase; and 3) according to the spectral curves, the types of ground objects are complex and diverse, with severe overlapping and mixture. Therefore, the datasets can show the effectiveness and generalization of the proposed method from different perspectives.

B. Feature Subset Classification

Since it is assumed that a feature subset with a larger amount of information and less redundancy has better classification advantages, the performance of the evolutionary algorithms can be evaluated by classification accuracy.

We used an SVM classifier, took 10% of the samples for each category as the training set, and used the rest of the samples as the test set to classify the subimages corresponding to the feature subsets. In the quantitative evaluation, the overall accuracy (OA) (19) and the kappa coefficient (Kappa) (20) obtained by the same SVM classifier are compared. The better the feature subset characteristics, the greater the value of the classification accuracy parameters mentioned above

$$\mathrm{OA} = \frac{\sum_{i=1}^{N_c} Na_i}{\sum_{i=1}^{N_c} N_i} \tag{19}$$

$$\mathrm{Kappa} = \frac{\sum_{i=1}^{N_c} N_i \cdot \sum_{i=1}^{N_c} Na_i - \sum_{i=1}^{N_c} \mathrm{Nc}_i \cdot N_i}{\left(\sum_{i=1}^{N_c} N_i\right)^2 - \sum_{i=1}^{N_c} \mathrm{Nc}_i \cdot N_i} \tag{20}$$

where Na_i and N_i represent the number of samples correctly classified and the total number of samples for the ith class, respectively. Nc_i refers to the number of samples classified into the ith class, and N_c corresponds to the number of categories in the image dataset.

The curve of the classification accuracy corresponding to each algorithm changes with the preset size of the subset, which was varied from 3 to 60 with a step size of 3, as shown in Fig. 9. The mean of the OA and Kappa for each algorithm, as well as their standard deviations and maximum values, is listed in Table II for quantitative evaluation.

In the Indian Pines dataset, the performance of the single-objective methods and the traditional methods are almost the same, while the multiobjective methods are significantly superior to them. However, this is not the case in the WHU-Hi-HanChuan dataset. The traditional clustering-based methods achieve higher accuracy than all the single-objective methods and most of the multiobjective ones. As for the WHU-Hi-HongHu and WHU-Hi-LongKou datasets, the feature subsets obtained by the whole evolutionary feature selection methods are more suitable for classification. Among them, multiobjective methods have slightly better than single-objective ones. In general, the proposed algorithm obtains the best average classification accuracy on all the hyperspectral remote sensing image datasets. Compared with its baseline algorithm, i.e., MOEA/D, MOCSA/D_FS obtains results with higher accuracy, revealing a stronger local search capability.

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TABLE II
CLASSIFICATION PERFORMANCE FOR THE DIFFERENT HYPERSPECTRAL REMOTE SENSING IMAGE DATASETS

Feature selection method	Indian Pines dataset				WHU-Hi-HanChuan dataset			
	Mean_OA±Std	Mean_K±Std	Max_OA	Max_K	Mean_OA±Std	Mean_K±Std	Max_OA	Max_K
1	0.7772	0.7444	0.7772	0.7444	0.8630	0.8394	0.8630	0.8394
2	0.7158±0.0803	0.6699±0.0985	0.7839	0.7522	0.7638±0.0802	0.7187±0.1050	0.8254	0.7949
3	0.7059±0.0765	0.6577±0.0979	0.7709	0.7375	0.7930±0.0582	0.7554±0.0726	0.8320	0.8026
4	0.7093±0.0678	0.6633±0.0843	0.7673	0.7331	**0.7946±0.0334**	**0.7581±0.0403**	0.8296	0.7997
5	0.7155±0.0750	0.6686±0.0938	0.7659	0.7306	0.7751±0.0816	0.7327±0.1062	**0.8326**	**0.8033**
6	0.6894±0.0776	0.6366±0.0981	0.7706	0.7362	0.7700±0.0854	0.7256±0.1100	0.8324	0.8030
7	0.7085±0.0725	0.6605±0.0904	0.7716	0.7371	0.7762±0.0844	0.7336±0.1107	**0.8337**	**0.8046**
8	0.7117±0.0701	0.6639±0.0887	0.7699	0.7351	0.7778±0.0659	0.7370±0.0830	0.8309	0.8013
9	0.7509±0.0700	0.7108±0.0865	**0.8096**	**0.7814**	0.7897±0.0586	0.7513±0.0730	0.8317	0.8023
10	0.7539±0.0579	0.7149±0.0718	0.8035	0.7746	0.7922±0.0519	0.7547±0.0643	0.8322	0.8029
11	0.7441±0.0602	0.7030±0.0756	0.7926	0.7616	0.7904±0.0560	0.7524±0.0700	0.8324	0.8032
12	0.7569±0.0601	0.7188±0.0723	0.8053	0.7762	0.7884±0.0610	0.7495±0.0775	0.8324	0.8031
13	**0.7578±0.0576**	**0.7192±0.0721**	0.8048	0.7759	0.7938±0.0507	0.7567±0.0626	0.8323	0.8029
14	**0.7652±0.0593**	**0.7279±0.0739**	**0.8113**	**0.7836**	**0.7981±0.0501**	**0.7616±0.0627**	0.8323	0.8029

Feature selection method	WHU-Hi-HongHu dataset				WHU-Hi-LongKou dataset			
	Mean_OA±Std	Mean_K±Std	Max_OA	Max_K	Mean_OA±Std	Mean_K±Std	Max_OA	Max_K
1	0.8466	0.8049	0.8466	0.8049	0.9885	0.9849	0.9885	0.9849
2	0.8021±0.0427	0.7452±0.0581	0.8371	0.7922	0.9570±0.0179	0.9432±0.0237	0.9770	0.9697
3	0.7719±0.0478	0.7065±0.0646	0.8106	0.7582	0.9461±0.0486	0.9283±0.0662	0.9716	0.9626
4	0.7604±0.0544	0.6899±0.0793	0.8059	0.7521	0.9426±0.0385	0.9240±0.0521	0.9705	0.9612
5	0.8114±0.0459	0.7571±0.0640	**0.8418**	**0.7983**	0.9685±0.0153	0.9585±0.0203	0.9804	0.9742
6	0.8080±0.0477	0.7526±0.0665	0.8380	0.7933	0.9685±0.0166	0.9585±0.0221	0.9806	0.9745
7	0.8109±0.0443	0.7580±0.0614	0.8405	0.7966	0.9687±0.0154	0.9587±0.0205	0.9802	0.9740
8	0.8112±0.0447	0.7571±0.0617	0.8408	0.7970	0.9677±0.0155	0.9574±0.0206	0.9806	0.9744
9	0.8127±0.0419	0.7593±0.0575	0.8397	0.7956	0.9677±0.0194	0.9574±0.0259	0.9811	0.9754
10	0.8119±0.0356	0.7584±0.0481	0.8379	0.7931	0.9695±0.0153	0.9598±0.0203	0.9821	0.9764
11	0.8141±0.0258	0.7615±0.0346	0.8350	**0.7984**	0.9710±0.0146	0.9618±0.0194	**0.9821**	**0.9764**
12	0.8126±0.0431	0.7590±0.0595	0.8369	0.7917	0.9704±0.0150	0.9610±0.0200	**0.9826**	**0.9771**
13	**0.8158±0.0305**	**0.7637±0.0410**	0.8391	0.7947	**0.9711±0.0146**	**0.9619±0.0193**	0.9819	0.9761
14	**0.8188±0.0304**	**0.7676±0.0411**	**0.8408**	0.7969	**0.9711±0.0148**	**0.9619±0.0196**	0.9817	0.9759

Notes: 1: all features. 2: DSEBS. 3: NC-OC-IE. 4: FNGBS. 5: GA. 6: DE. 7: PSO. 8: SCA. 9: NSGA-II. 10: NSGA-III. 11: MOPSO. 12: MOSCA*. 13: MOEA/D. 14: MOCSA/D_FS.

C. Multiobjective Pareto Front

Since the multiobjective optimization algorithms, finally, obtain a Pareto solution set, their corresponding PF in objective space can express the global and local capabilities of the algorithm to some extent. The experiment described in this section was conducted to verify the improvement by contrasting the actual distribution of the PFs optimized by the different methods and their parametric characteristics. The algorithm parameters and the preset size of the feature subset were the same as those described in Section IV-A.

For the maximization problem modeled in this article, the rules can be summarized for judging the distribution of the PF as follows. The closer the PF surface is to the upper right, the higher the degree of optimization of the algorithm; the longer the PF surface curve, the more extensive the solution distribution; and the farther away the points on the PF surface, the greater the difference between the solutions.

What is more, we adopted the characteristic parameters of the PF surface for a quantitative comparison [51], i.e., the hypervolume (HV) (21), pure diversity (PD) (22), and spacing (S) (23). HV expresses the area of the range dominated by the PF and characterizes the distribution diversity and convergence of the PF. PD evaluates the distribution uniformity of the PF by analyzing the difference between the nondominant solutions and the population composed of other solutions in turn. S is the standard deviation of the minimum Euclidean distance between the PF points. Therefore, the larger the values of HV and PD, the smaller S is, and the better the characteristics of the PF are

$$\mathrm{HV} = \mathrm{hypervolume}\left(\bigcup_{i=1}^{|\mathrm{PF}|} G_i\right) \tag{21}$$

$$\mathrm{PD}(s^*) = \max_{s_i \in s^*}\left(\mathrm{PD}(s^* - s_i) + d(s_i, s^* - s_i)\right)$$

$$d(s, s^*) = \max_{s \in s^*}\left(\mathrm{dissimilarity}(s, s^*)\right) \tag{22}$$

$$\mathrm{DisMat} = \mathrm{pdist2(PF, PF)}$$

$$S = \mathrm{std(min(DisMat, [], 2))} \tag{23}$$

where G_i is the position of the solution in the objective space and |PF| is the total number of solutions in the Pareto set. s_i represents the Pareto solution individuals, and $d(s, s^*)$ measures the difference between individual s and group s^*. DisMat is a matrix of the Euclidean distances between individuals, with the diagonal position (its own distance) set to infinity.

Fig. 10 demonstrates the distribution of Pareto solution sets of the various multiobjective methods, while Tables III–VI list the parameters related to the PF. In the Indian Pines dataset, the PFs of multiobjective methods gradually separate with the increase of the preset size of the feature subset. Decomposition-based methods obtain wider distribution with more diverse solutions. As for the WHU-Hi-HanChuan dataset, MOCSA/D_FS can always develop better PF, which

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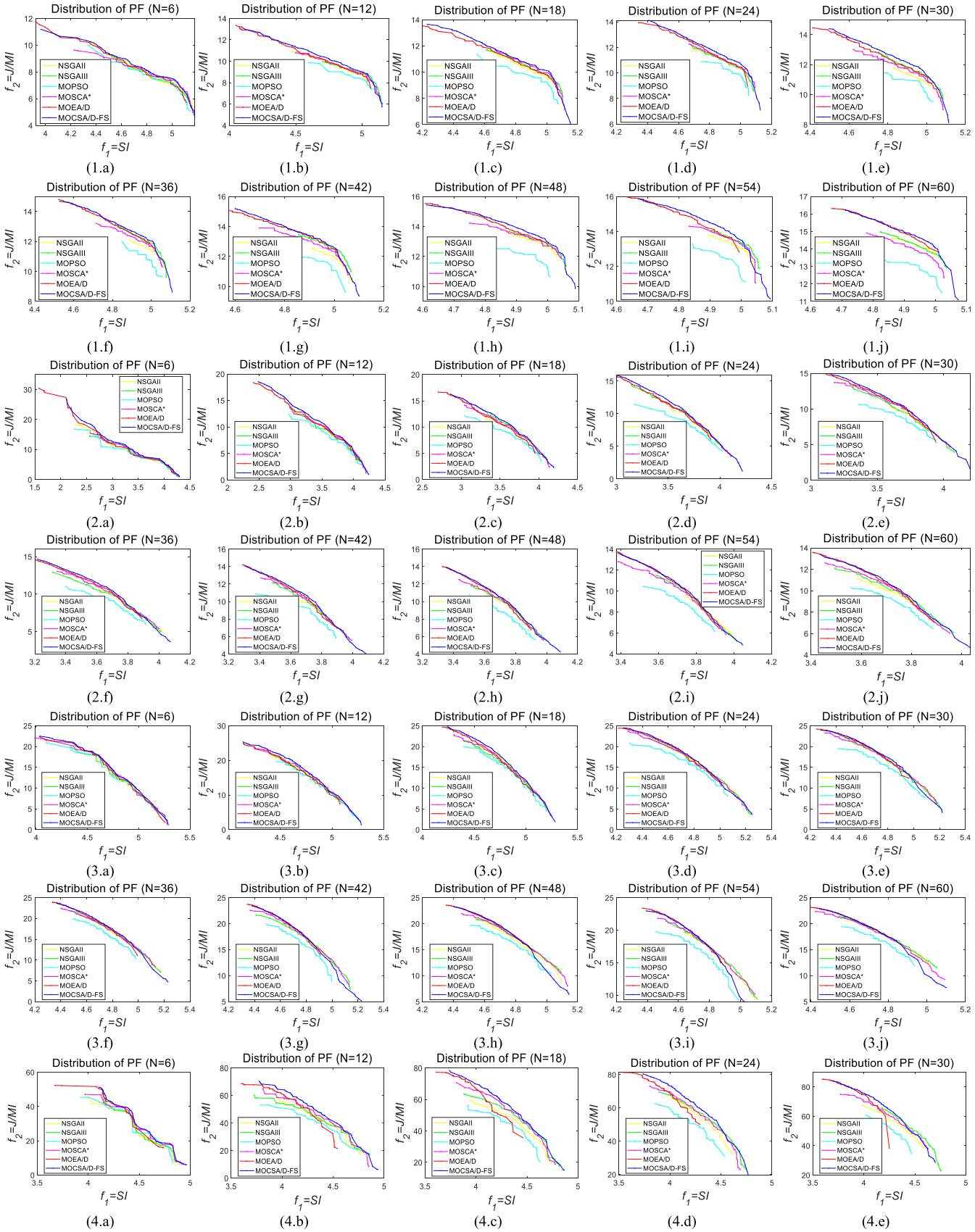
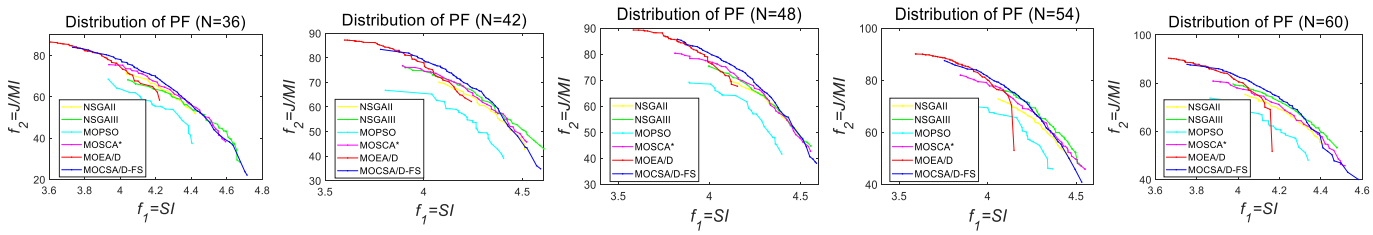


Fig. 10. PFs obtained by the different multiobjective feature selection methods under a preset size of feature subset for the four hyperspectral remote sensing image datasets. (1) Indian Pines dataset. (1.a)–(1.j) Preset size of the feature subset is 6–60, with a step size of 6. (2) WHU-Hi-HanChuan dataset. (3) WHU-Hi-LongKou dataset. (4) WHU-Hi-LongKou dataset. (2.a)–(2.j) Preset size of the feature subset is 6–60, with a step size of 6. (3.a)–(3.j) Preset size of the feature subset is 6–60, with a step size of 6. (4.a)–(4.j) Preset size of feature subset is 6–60, with a step size of 6.

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(4.f) (4.g) (4.h) (4.i) (4.j)

Fig. 10. *(Continued.)* PFs obtained by the different multiobjective feature selection methods under a preset size of feature subset for the four hyperspectral remote sensing image datasets. (1) Indian Pines dataset. (1.a)–(1.j) Preset size of the feature subset is 6–60, with a step size of 6. (2) WHU-Hi-HanChuan dataset. (3) WHU-Hi-LongKou dataset. (4) WHU-Hi-LongKou dataset. (2.a)–(2.j) Preset size of the feature subset is 6–60, with a step size of 6. (3.a)–(3.j) Preset size of the feature subset is 6–60, with a step size of 6. (4.a)–(4.j) Preset size of feature subset is 6–60, with a step size of 6.

TABLE III

MULTIOBJECTIVE PF PARAMETERS FOR THE INDIAN PINES DATASET

N	HV ↑						PD ↑						S ↓					
	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6
3	1.82e0	3.18e0	4.52e0	4.56e0	**4.82e0**	4.55e0	1.29e3	1.71e3	2.22e3	2.16e3	3.82e3	**3.94e3**	1.01e-1	1.24e-1	6.25e-2	6.22e-2	3.87e-2	**3.44e-2**
6	3.89e0	2.52e0	3.87e0	4.26e0	6.84e0	**7.59e0**	2.01e3	1.82e3	1.87e3	1.95e3	**4.28e3**	3.83e3	1.73e-1	8.73e-2	9.18e-2	6.25e-2	4.25e-2	**2.81e-2**
9	1.69e0	3.10e0	3.56e0	5.41e0	6.18e0	**7.58e0**	1.28e3	1.81e3	1.65e3	3.23e3	3.69e3	**4.18e3**	6.94e-2	5.44e-2	9.74e-2	7.77e-2	3.39e-2	**2.56e-2**
12	1.60e0	1.74e0	1.23e0	2.77e0	5.74e0	**7.63e0**	1.32e3	1.57e3	9.31e2	1.80e3	3.89e3	**4.01e3**	1.05e-1	7.76e-2	4.30e-2	5.49e-2	7.18e-2	**2.23e-2**
15	9.44e-1	1.65e0	1.68e0	3.18e0	5.33e0	**6.82e0**	1.06e3	1.92e3	9.63e2	2.14e3	2.91e3	**5.20e3**	9.31e-2	6.88e-2	8.16e-2	5.50e-2	4.03e-2	**1.93e-2**
18	1.66e0	1.77e0	1.95e0	2.03e0	4.13e0	**6.77e0**	1.27e3	1.75e3	1.40e3	1.79e3	3.12e3	**3.77e3**	9.13e-2	3.34e-2	1.19e-1	6.71e-2	3.10e-2	**2.10e-2**
21	1.11e0	1.14e0	1.20e0	2.10e0	3.37e0	**5.72e0**	1.34e3	1.31e3	1.02e3	1.90e3	2.32e3	**3.11e3**	4.67e-2	4.52e-2	1.10e-1	6.43e-2	4.72e-2	**2.62e-2**
24	9.22e-1	1.43e0	7.97e-1	1.43e0	3.44e0	**4.93e0**	1.12e3	1.29e3	6.97e2	1.36e3	2.45e3	**3.53e3**	7.71e-2	5.44e-2	1.25e-1	6.38e-2	3.89e-2	**2.53e-2**
27	3.60e-1	5.19e-1	9.31e-1	2.24e0	2.75e0	**4.40e0**	6.32e2	9.21e2	7.80e2	1.55e3	2.69e3	**3.10e3**	3.08e-2	4.18e-2	7.43e-2	5.78e-2	2.06e-2	**1.28e-2**
30	4.98e-1	1.03e0	6.93e-1	1.84e0	2.23e0	**3.80e0**	8.47e2	1.54e3	7.16e2	2.02e3	1.80e3	**3.18e3**	3.83e-2	5.52e-2	4.77e-2	4.66e-2	2.92e-2	**1.82e-2**
33	1.01e0	4.82e-1	7.19e-1	1.25e0	2.21e0	**4.63e0**	1.20e3	9.56e2	7.57e2	1.40e3	2.57e3	**3.39e3**	6.89e-2	4.18e-2	6.32e-2	4.16e-2	4.58e-2	**2.35e-2**
36	3.46e-1	9.22e-1	4.84e-1	9.84e-1	1.58e0	**3.47e0**	6.98e2	1.07e3	5.61e2	1.07e3	1.94e3	**2.27e3**	2.70e-2	9.64e-2	1.05e-1	3.20e-2	**2.02e-2**	2.06e-2
39	4.31e-1	1.20e0	5.89e-1	1.19e0	1.69e0	**3.12e0**	6.50e2	1.22e3	6.87e2	1.27e3	1.82e3	**2.40e3**	3.31e-2	5.37e-2	7.52e-2	3.35e-2	2.69e-2	**1.92e-2**
42	3.59e-1	5.22e-1	6.27e-1	1.12e0	1.33e0	**2.89e0**	7.33e2	8.29e2	6.77e2	1.01e3	1.73e3	**2.29e3**	4.29e-2	3.68e-2	1.59e-1	5.68e-2	**1.87e-2**	1.93e-2
45	3.65e-1	6.45e-1	4.76e-1	1.60e0	1.64e0	**2.44e0**	6.32e2	9.60e2	5.36e2	1.23e3	**2.06e3**	1.93e3	3.49e-2	3.56e-2	4.92e-2	9.53e-2	5.41e-2	**2.87e-2**
48	2.52e-1	4.21e-1	4.74e-1	6.86e-1	1.12e0	**2.64e0**	5.47e2	7.06e2	4.93e2	8.86e2	1.91e3	**2.70e3**	4.03e-2	3.94e-2	3.17e-2	2.77e-2	**2.04e-2**	2.94e-2
51	3.61e-1	7.38e-1	7.30e-1	8.77e-1	1.21e0	**1.90e0**	6.06e2	1.17e3	7.26e2	9.42e2	**2.06e3**	1.67e3	6.51e-2	4.34e-2	6.55e-2	3.38e-2	2.84e-2	**1.14e-2**
54	2.67e-1	3.36e-1	3.65e-1	6.88e-1	1.12e0	**2.40e0**	5.67e2	6.62e2	5.52e2	8.11e2	1.51e3	**3.00e3**	3.00e-2	3.44e-2	6.85e-2	3.76e-2	5.83e-2	**1.56e-2**
57	2.76e-1	2.92e-1	3.21e-1	1.06e0	1.02e0	**2.05e0**	5.91e2	5.93e2	5.22e2	1.17e3	1.48e3	**2.04e3**	6.49e-2	2.75e-2	1.15e-1	4.86e-2	2.69e-2	**1.61e-2**
60	1.51e-1	4.28e-1	3.52e-1	6.31e-1	8.51e-1	**1.87e0**	6.00e2	7.88e2	4.65e2	9.94e2	1.33e3	**2.05e3**	6.81e-2	3.55e-2	1.16e-1	4.77e-2	**2.30e-2**	2.96e-2
Ave	9.15e-1	1.20e0	1.28e0	2.00e0	2.93e0	**4.36e0**	9.50e2	1.23e3	9.12e2	1.53e3	2.47e3	**3.08e3**	6.50e-2	5.44e-2	8.50e-2	5.33e-2	3.58e-2	**2.23e-2**

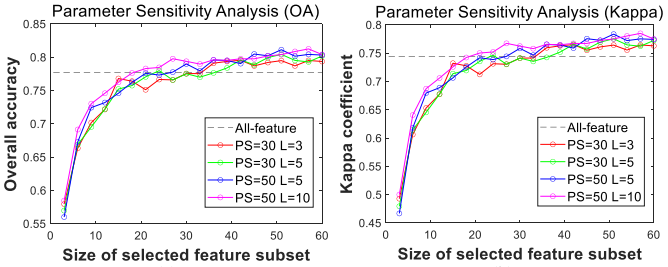
Notes: N: preset size of feature subset. M1: NSGA-II. M2: NSGA-III. M3: MOPSO. M4: MOSCA*. M5: MOEA/D. M6: MOCSA/D_FS. Ave: parameter average for the different preset size of feature subset.

is closer to the upper right in the objective space. Although the PFs optimized by the proposed method for the last two datasets have certain disadvantages in the lower right, the solutions are distributed more uniformly to be ideal surfaces. The optimization degree of the knee point is higher. Besides, the parameters of PF obtained by MOCSA/D_FS are at the optimal level in most cases. Thus, it can be considered that the proposed method can better optimize the multiobjective feature selection problem.

D. Parameter Sensitivity Analysis

In order to test the robustness of MOCSA/D_FS, a parameter sensitivity analysis was performed. The population size PS determines the number of subproblems for the decomposition, while the neighborhood size L is closely related to the process of clonal selection. As a result, these two parameters are sensitive parameters, and they affect the global and local optimization ability of the algorithm.

PS and L are modified to {30, 30, 50, 50} and {3, 5, 5, 10}, respectively, with the other parameters remaining. The Indian Pines dataset was selected for the feature selection.



(a) (b)

Fig. 11. Classification accuracy curves for different parameter settings for MOCSA/D_FS on the Indian Pines dataset. (a) Overall accuracy. (b) Kappa coefficient.

As shown in Fig. 11 and Table VII, the obtained curve is the highest when the neighborhood is larger ($L = 10$), and the classification effect is better when the preset size of features is larger. The first two rows and the last two rows in the analysis table show that, for the same PS, the change of L has a great impact on the maximum classification accuracy, i.e., it affects the local search ability. For the same L, but with different PS (the second and third rows in Table VII),

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TABLE IV
MULTIOBJECTIVE PF PARAMETERS FOR THE WHU-HI-HANCHUAN DATASET

N	HV ↑						PD ↑						S ↓					
	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6
3	2.19e0	5.19e1	1.80e2	2.12e2	2.00e2	**2.15e2**	1.64e3	9.04e3	1.90e4	1.90e4	**2.34e4**	2.26e4	**1.89e-1**	6.21e-1	1.80e0	1.24e0	1.60e0	1.30e0
6	2.76e1	1.94e1	2.45e1	2.09e1	**5.23e1**	4.11e1	6.36e3	5.44e3	5.55e3	5.11e3	**1.54e4**	9.62e3	4.18e-1	2.95e-1	1.73e-1	1.44e-1	2.02e-1	**6.59e-2**
9	9.47e0	2.72e1	1.07e1	2.12e1	2.67e1	**3.28e1**	3.33e3	7.79e3	4.24e3	6.57e3	9.00e3	**9.05e3**	2.77e-1	5.30e-1	1.91e-1	1.45e-1	1.59e-1	**2.96e-2**
12	1.01e1	1.32e1	1.01e1	1.33e1	2.14e1	**2.58e1**	4.13e3	4.99e3	3.77e3	5.21e3	**8.45e3**	7.99e3	1.56e-1	1.61e-1	1.80e-1	1.51e-1	8.07e-2	**2.61e-2**
15	1.34e1	9.10e0	7.88e0	1.18e1	1.52e1	**2.12e1**	4.07e3	4.49e3	3.88e3	4.98e3	7.12e3	**7.15e3**	3.40e-1	2.01e-1	1.74e-1	1.31e-1	8.16e-2	**4.02e-2**
18	7.91e0	6.43e0	8.00e0	1.19e1	1.33e1	**1.67e1**	3.26e3	3.05e3	3.15e3	4.29e3	6.07e3	**8.00e3**	1.66e-1	1.30e-1	1.23e-1	1.66e-1	5.27e-2	**2.17e-2**
21	4.31e0	6.18e0	4.78e0	6.21e0	1.13e1	**1.66e1**	2.26e3	2.97e3	2.53e3	2.99e3	4.59e3	**8.68e3**	9.36e-2	1.50e-1	1.00e-1	9.24e-2	1.06e-1	**2.29e-2**
24	4.46e0	8.01e0	5.80e0	6.38e0	8.31e0	**1.59e1**	2.54e3	4.29e3	2.69e3	2.94e3	**4.72e3**	4.72e3	8.54e-2	1.20e-1	1.71e-1	8.44e-2	4.88e-2	**1.95e-2**
27	3.45e0	5.03e0	5.53e0	5.74e0	7.52e0	**1.35e1**	2.55e3	2.76e3	2.69e3	3.27e3	4.56e3	**7.28e3**	8.88e-2	9.27e-2	1.27e-1	7.43e-2	4.71e-2	**1.89e-2**
30	2.82e0	6.63e0	3.08e0	7.43e0	7.35e0	**1.29e1**	2.12e3	3.41e3	2.13e3	3.13e3	4.03e3	**5.43e3**	7.59e-2	1.06e-1	7.40e-2	9.23e-2	7.31e-2	**2.20e-2**
33	2.49e0	4.19e0	3.25e0	6.51e0	6.30e0	**1.07e1**	2.57e3	3.34e3	2.06e3	3.22e3	5.21e3	**5.86e3**	7.06e-2	9.63e-2	9.82e-2	1.09e-1	4.81e-2	**1.70e-2**
36	2.36e0	5.25e0	2.50e0	4.82e0	5.16e0	**8.85e0**	1.92e3	3.36e3	1.73e3	2.86e3	3.54e3	**4.77e3**	1.01e-1	1.33e-1	8.99e-2	9.04e-2	4.95e-2	**2.26e-2**
39	3.14e0	4.51e0	2.55e0	5.44e0	4.53e0	**8.12e0**	2.20e3	3.55e3	1.82e3	3.22e3	3.25e3	**6.33e3**	8.49e-2	6.77e-2	8.01e-2	1.30e-1	3.51e-2	**1.66e-2**
42	1.13e0	3.11e0	2.60e0	4.03e0	4.04e0	**7.33e0**	1.15e3	2.58e3	1.79e3	2.43e3	3.52e3	**4.58e3**	3.88e-2	7.09e-2	6.42e-2	8.08e-2	3.10e-2	**1.45e-2**
45	1.90e0	3.45e0	2.39e0	3.62e0	3.76e0	**7.67e0**	1.73e3	2.54e3	1.99e3	2.54e3	3.04e3	**3.95e3**	1.06e-1	8.07e-2	6.78e-2	7.70e-2	3.32e-2	**2.24e-2**
48	2.19e0	2.95e0	2.17e0	3.66e0	4.01e0	**6.74e0**	2.00e3	2.22e3	1.77e3	2.20e3	3.50e3	**5.37e3**	1.05e-1	5.22e-2	9.09e-2	7.30e-2	3.37e-2	**1.45e-2**
51	2.61e0	2.66e0	2.26e0	3.92e0	3.79e0	**5.60e0**	1.75e3	2.19e3	1.54e3	2.81e3	2.86e3	**4.10e3**	6.36e-2	9.32e-2	2.23e-1	9.89e-2	3.87e-2	**1.68e-2**
54	8.80e-1	2.59e0	1.61e0	3.86e0	3.88e0	**5.53e0**	1.28e3	2.10e3	1.59e3	2.80e3	3.09e3	**4.65e3**	4.21e-2	7.53e-2	8.61e-2	9.34e-2	3.30e-2	**1.33e-2**
57	1.16e0	2.50e0	1.66e0	3.28e0	2.83e0	**5.21e0**	1.26e3	2.27e3	1.43e3	2.16e3	2.88e3	**3.66e3**	5.44e-2	7.40e-2	6.23e-2	8.39e-2	2.44e-2	**1.16e-2**
60	1.80e0	2.77e0	1.24e0	3.17e0	3.07e0	**5.03e0**	1.63e3	2.03e3	1.29e3	2.32e3	3.10e3	**4.27e3**	8.29e-2	5.57e-2	4.27e-2	5.86e-2	3.06e-2	**1.45e-2**
Ave	5.27e0	9.36e0	1.42e1	1.79e1	2.02e1	**2.41e1**	2.49e3	3.72e3	3.33e3	4.20e3	6.07e3	**6.90e3**	1.32e-1	1.60e-1	2.01e-1	1.61e-1	1.41e-1	**8.64e-2**

TABLE V
MULTIOBJECTIVE PF PARAMETERS FOR THE WHU-HI-HONGHU DATASET

N	HV ↑						PD ↑						S ↓					
	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6
3	2.21e1	2.51e1	2.20e1	2.59e1	**2.62e1**	2.58e1	5.71e3	6.06e3	6.56e3	6.84e3	**9.13e3**	8.72e3	3.08e-1	4.14e-1	9.84e-2	2.89e-1	1.01e-1	**3.68e-2**
6	8.41e0	1.55e1	2.02e1	2.46e1	2.33e1	**2.47e1**	3.16e3	4.94e3	6.16e3	6.03e3	7.33e3	**7.57e3**	2.29e-1	3.47e-1	2.18e-1	2.11e-1	1.07e-1	**2.98e-2**
9	1.48e1	1.16e1	1.82e1	1.94e1	2.14e1	**2.40e1**	4.71e3	4.11e3	6.71e3	5.45e3	7.74e3	**8.65e3**	3.40e-1	2.24e-1	2.42e-1	2.16e-1	1.44e-1	**4.10e-2**
12	1.19e1	1.53e1	1.14e1	1.91e1	1.55e1	**2.54e1**	4.19e3	5.08e3	4.19e3	5.33e3	6.86e3	**8.50e3**	2.83e-1	3.69e-1	1.81e-1	2.45e-1	1.36e-1	**4.12e-2**
15	1.13e1	1.70e1	1.00e1	1.79e1	1.98e1	**2.42e1**	4.14e3	5.87e3	5.21e3	5.99e3	**7.00e3**	6.89e3	1.88e-1	1.58e-1	2.32e-1	2.25e-1	9.65e-2	**3.97e-2**
18	9.85e0	1.46e1	1.19e1	1.71e1	1.39e1	**2.21e1**	4.19e3	5.15e3	4.78e3	5.09e3	6.08e3	**9.23e3**	1.41e-1	2.06e-1	1.56e-1	2.11e-1	1.04e-1	**6.40e-2**
21	8.31e0	1.52e1	9.39e0	1.67e1	1.40e1	**2.16e1**	3.33e3	5.03e3	4.24e3	5.05e3	**8.15e3**	7.87e3	2.94e-1	3.21e-1	2.15e-1	2.49e-1	1.50e-1	**4.02e-2**
24	7.80e0	1.36e1	9.36e0	1.64e1	1.07e1	**1.95e1**	5.10e3	5.90e3	3.59e3	4.72e3	5.83e3	**6.58e3**	1.60e-1	2.59e-1	1.86e-1	2.73e-1	7.92e-2	**4.94e-2**
27	9.94e0	1.10e1	7.16e0	1.64e1	1.18e1	**1.86e1**	4.05e3	4.58e3	3.38e3	5.15e3	5.83e3	**6.74e3**	2.64e-1	1.96e-1	1.15e-1	2.68e-1	9.54e-2	**3.88e-2**
30	8.82e0	1.22e1	7.74e0	1.40e1	8.21e0	**1.77e1**	3.65e3	4.76e3	3.59e3	4.75e3	4.85e3	**7.39e3**	2.39e-1	3.44e-1	1.24e-1	1.92e-1	8.16e-2	**3.86e-2**
33	5.82e0	1.12e1	6.08e0	1.26e1	6.93e0	**1.64e1**	3.64e3	4.80e3	3.60e3	4.32e3	4.68e3	**7.20e3**	1.30e-1	1.54e-1	1.19e-1	1.43e-1	5.22e-2	**3.56e-2**
36	7.77e0	1.01e1	4.58e0	9.78e0	8.24e0	**1.55e1**	4.65e3	4.11e3	3.26e3	4.53e3	4.73e3	**8.22e3**	2.25e-1	2.00e-1	1.18e-1	1.63e-1	1.21e-1	**4.60e-2**
39	5.38e0	8.95e0	5.04e0	1.19e1	6.10e0	**1.48e1**	3.20e3	3.73e3	2.80e3	4.52e3	3.73e3	**6.20e3**	3.16e-1	1.63e-1	1.22e-1	2.09e-1	5.27e-2	**4.55e-2**
42	3.58e0	9.98e0	5.44e0	1.07e1	5.41e0	**1.46e1**	2.45e3	4.63e3	3.39e3	4.30e3	4.30e3	**6.39e3**	8.40e-2	1.72e-1	2.16e-1	1.92e-1	7.12e-2	**2.33e-2**
45	5.50e0	1.06e1	4.53e0	9.39e0	8.13e0	**1.31e1**	3.79e3	4.60e3	2.59e3	4.24e3	4.69e3	**6.43e3**	1.39e-1	1.94e-1	1.04e-1	1.64e-1	1.11e-1	**3.54e-2**
48	6.54e0	5.53e0	4.73e0	9.23e0	5.02e0	**1.21e1**	3.19e3	3.27e3	3.27e3	3.91e3	3.43e3	**5.67e3**	4.30e-1	1.39e-1	8.31e-2	1.58e-1	5.41e-2	**2.97e-2**
51	5.15e0	7.66e0	4.30e0	1.01e1	3.49e0	**1.12e1**	3.75e3	3.32e3	2.57e3	4.19e3	2.82e3	**5.93e3**	2.44e-1	9.74e-2	1.71e-1	1.91e-1	3.87e-2	**2.99e-2**
54	4.57e0	6.86e0	5.48e0	7.11e0	4.23e0	**8.59e0**	2.78e3	**5.45e3**	2.77e3	4.06e3	3.27e3	4.62e3	2.46e-1	1.60e-1	3.60e-1	1.28e-1	5.34e-2	**2.33e-2**
57	3.98e0	4.76e0	5.21e0	7.99e0	4.00e0	**1.23e1**	3.26e3	3.86e3	3.33e3	3.79e3	3.57e3	**5.62e3**	2.02e-1	1.04e-1	2.14e-1	1.23e-1	6.73e-2	**3.41e-2**
60	4.08e0	4.94e0	2.79e0	8.54e0	2.53e0	**9.69e0**	2.62e3	3.38e3	1.93e3	4.12e3	2.57e3	**6.10e3**	1.09e-1	8.34e-2	1.90e-1	1.27e-1	**3.25e-2**	4.72e-2
Ave	8.28e0	1.16e1	8.77e0	1.42e1	1.10e1	**1.76e1**	3.78e3	4.63e3	3.90e3	4.82e3	5.33e3	**7.03e3**	2.29e-1	2.15e-1	1.73e-1	1.99e-1	8.75e-2	**3.85e-2**

the more detailed decomposition of the subproblems also improves the classification accuracy. This is mainly because, in the search space of the same size, the number of subproblems increases, representing a narrowing of the search scope assigned by each individual, which is conducive to local development.

E. Computational Complexity Analysis

For the computational complexity of the proposed method, it is assumed that there are N_A features and $N_P = M \times N$ pixels in the original hyperspectral image. The maximum of the iteration and the size of the neighborhood are denoted as $T_{\max}$ and L, respectively. Thus, there are three main steps for the computational complexity as follows.

1) Initialization of the Population: Since there are PS solutions in a population and N_A variables in one solution, the computational complexity is $O(\mathrm{PS} \times N_A)$.

2) Calculation of the Objective Functions and Extraction and Update of Nondominant Sets: It's assumed that N_B features are selected from the whole features, and there are N_P pixels in each feature.

1) N_B features are selected for calculation of the value of the functions, so the computational complexity for f_1 is $O_1 = O(\mathrm{PS} \times N_B \times N_P)$ and for f_2 is $O_2 = O(\mathrm{PS} \times C_{N_B}^2 \times N_P)$.
2) To extract and update the nondominant sets, comparisons for the two objective functions in a population are needed. Thus, the computational complexity for extract

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TABLE VI
MULTIOBJECTIVE PF PARAMETERS FOR THE WHU-HI-LONGKOU DATASET

N	HV ↑						PD ↑						S ↓					
	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6	M1	M2	M3	M4	M5	M6
3	2.56e1	3.15e1	2.68e1	2.87e1	**3.43e1**	3.40e1	6.62e3	6.64e3	5.84e3	6.96e3	1.04e4	**1.37e4**	5.69e-1	8.98e-1	3.13e-1	3.76e-1	2.45e-1	**1.61e-1**
6	3.45e1	2.74e1	3.27e1	**3.69e1**	3.67e1	3.58e1	8.23e3	6.83e3	5.49e3	7.52e3	7.49e3	**1.22e4**	4.07e-1	1.18e0	1.09e0	3.75e-1	6.98e-1	**2.19e-1**
9	4.80e1	3.07e1	3.20e1	5.49e1	4.83e1	**6.02e1**	1.15e4	7.00e3	5.92e3	**1.30e4**	1.24e4	1.28e4	1.08e0	1.59e0	7.19e-1	1.44e0	7.38e-1	**1.18e-1**
12	3.14e1	4.57e1	3.00e1	5.45e1	4.26e1	**6.85e1**	8.21e3	8.98e3	6.16e3	1.29e4	1.06e4	**1.40e4**	5.48e-1	3.74e-1	7.00e-1	6.92e-1	6.96e-1	**2.09e-1**
15	3.58e1	3.37e1	2.56e1	5.04e1	4.14e1	**6.41e1**	9.11e3	7.06e3	6.06e3	9.73e3	9.69e3	**1.28e4**	8.53e-1	4.62e-1	8.17e-1	5.59e-1	6.27e-1	**3.35e-1**
18	3.09e1	4.08e1	2.45e1	4.75e1	3.41e1	**6.61e1**	7.46e3	8.44e3	6.28e3	8.86e3	8.08e3	**1.37e4**	6.14e-1	4.65e-1	9.42e-1	6.01e-1	8.50e-1	**1.94e-1**
21	2.82e1	3.64e1	2.67e1	4.37e1	2.44e1	**5.99e1**	1.10e4	8.96e3	5.95e3	9.75e3	8.92e3	**1.54e4**	5.19e-1	8.92e-1	7.46e-1	4.57e-1	4.46e-1	**2.06e-1**
24	1.96e1	3.96e1	2.08e1	4.25e1	2.29e1	**5.94e1**	1.05e4	9.46e3	5.52e3	9.19e3	6.78e3	**1.31e4**	4.04e-1	6.76e-1	9.64e-1	5.91e-1	4.32e-1	**1.83e-1**
27	2.32e1	2.55e1	1.45e1	4.57e1	2.98e1	**5.23e1**	8.66e3	7.28e3	3.80e3	7.94e3	7.12e3	**1.13e4**	4.10e-1	3.96e-1	4.22e-1	5.32e-1	1.21e0	**1.76e-1**
30	1.74e1	2.43e1	1.16e1	3.08e1	3.03e1	**5.35e1**	7.43e3	6.71e3	3.95e3	9.56e3	9.69e3	**1.12e4**	5.66e-1	5.72e-1	6.04e-1	3.67e-1	2.19e0	**1.73e-1**
33	2.17e1	1.82e1	1.22e1	3.04e1	1.84e1	**6.05e1**	6.77e3	6.07e3	3.88e3	7.01e3	6.48e3	**1.35e4**	5.55e-1	4.09e-1	1.33e0	5.83e-1	5.42e-1	**1.32e-1**
36	7.98e0	2.46e1	1.45e1	2.39e1	1.70e1	**5.78e1**	4.13e3	6.33e3	4.16e3	6.14e3	6.44e3	**1.44e4**	3.32e-1	5.98e-1	9.57e-1	4.65e-1	5.20e-1	**1.91e-1**
39	1.14e1	2.79e1	1.43e1	2.74e1	2.51e1	**4.25e1**	5.04e3	8.40e3	3.90e3	6.98e3	6.19e3	**1.28e4**	4.77e-1	5.83e-1	5.68e-1	6.73e-1	2.83e0	**1.61e-1**
42	1.18e1	2.27e1	1.64e1	1.91e1	1.58e1	**3.81e1**	5.55e3	7.82e3	4.03e3	6.29e3	6.25e3	**1.30e4**	6.30e-1	4.52e-1	6.31e-1	3.31e-1	3.24e-1	**1.70e-1**
45	1.46e1	1.46e1	1.63e1	2.83e1	2.70e1	**4.43e1**	7.62e3	5.50e3	3.76e3	8.20e3	8.09e3	**1.38e4**	6.44e-1	3.43e-1	1.21e0	4.05e-1	1.42e0	**1.08e-1**
48	8.39e0	1.68e1	1.39e1	2.73e1	1.22e1	**3.52e1**	4.93e3	5.63e3	3.94e3	7.64e3	5.64e3	**1.27e4**	2.38e-1	4.66e-1	9.69e-1	5.06e-1	2.26e-1	**1.50e-1**
51	6.94e0	1.71e1	7.28e0	2.21e1	1.70e1	**3.46e1**	5.68e3	5.54e3	2.92e3	5.86e3	5.91e3	**1.37e4**	2.28e-1	4.51e-1	6.58e-1	5.13e-1	4.30e-1	**9.91e-2**
54	9.63e0	1.23e1	1.36e1	2.48e1	2.05e1	**3.51e1**	4.17e3	5.08e3	3.73e3	8.07e3	6.43e3	**1.04e4**	4.01e-1	3.81e-1	7.48e-1	3.72e-1	2.27e0	**1.74e-1**
57	1.36e1	1.70e1	8.56e0	2.02e1	1.24e1	**3.35e1**	6.82e3	6.37e3	2.94e3	5.83e3	5.65e3	**1.27e4**	5.66e-1	4.71e-1	3.17e-1	3.57e-1	4.07e-1	**1.88e-1**
60	7.86e0	1.31e1	1.21e1	2.19e1	1.92e1	**3.83e1**	4.46e3	6.69e3	3.65e3	6.81e3	6.62e3	**1.01e4**	2.59e-1	5.30e-1	7.64e-1	4.19e-1	2.16e0	**1.66e-1**
Ave	2.04e1	2.60e1	1.87e1	3.41e1	2.65e1	**4.87e1**	7.19e3	7.04e3	4.59e3	8.21e3	7.74e3	**1.29e4**	5.15e-1	6.09e-1	7.74e-1	5.31e-1	9.64e-1	**1.76e-1**

TABLE VII
CLASSIFICATION PERFORMANCE WITH DIFFERENT PARAMETER SETTINGS FOR MOCSA/D_FS ON THE INDIAN PINES DATASET

PS	L	Mean_OA±Std	Mean_K±Std	Max_OA	Max_K
30	3	0.7582±0.0547	0.7199±0.0679	0.7955	0.7651
30	5	0.7583±0.0569	0.7201±0.0708	0.8043	0.7754
50	5	0.7652±0.0593	0.7279±0.0739	0.8113	0.7836
50	10	0.7729±0.0534	0.7373±0.0661	0.8125	0.7850

and update is considered as $O_3 = O(2 \times C_{\mathrm{PS}}^2)$ and $O_4 = O(2 \times \mathrm{PS})$.

3) Main Loop of the Decomposition-Based Multiobjective Clonal Selection Algorithm: $T_{\max}$ iterations are executed for the best feature subset.

1) With operations on the whole objective function vectors, the computational complexity for clone amplification is $O(4 \times T_{\max} \times \mathrm{PS} \times L)$ and for hypermutation is $O(T_{\max} \times \mathrm{PS} \times L \times N_A)$. Then, the evaluation for the obtained solutions is $O(T_{\max} \times (O_1 + O_2) \times L)$.
2) After the generation of new solutions, the computational complexity for the update of the nondominated sets and the selection of the best individuals is $O(T_{\max} \times O_4)$ and $O(T_{\max} \times \mathrm{PS} \times L)$.

V. CONCLUSION

In this article, in view of the shortcomings of the traditional MOEA/D when applied to feature selection, a novel normalized decomposition method has been proposed, and the idea of clonal selection was combined to make up an effective decomposition-based multiobjective CSA with balanced global and local optimization capabilities. In the MOCSA/D_FS, two critical metrics—the ratio between the relative scatter value and the MI (J/MI), together with the SI—are introduced to measure the redundancy and information of the solutions in the feature selection problem. This multiobjective problem is then optimized using the proposed method. Since the result is a Pareto solution set, a knee-point selection strategy is applied to select the optimal solution on the PF surface. Finally, the SVM classifier is used to classify the feature subset image. To fully verify the effectiveness and generalizability of the proposed method, experiments were performed on four hyperspectral remote sensing image datasets. The results showed that MOCSA/D_FS can achieve the optimal feature subset output under the optimization of the information amount and redundancy tradeoff.

In future work, evaluation criteria more suitable for the hyperspectral image feature selection problem should be further studied. What is more, given the strong correlation between features, some more productive evolutionary operators expected to reduce the randomness of the evolution should be researched to improve the solving efficiency.

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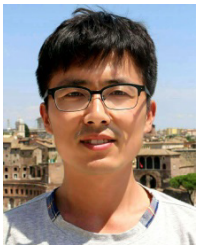
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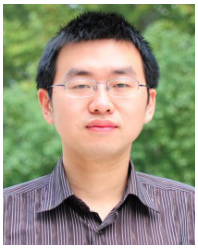
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